

SS1 Mathematics Diagnostic: Cambridge IGCSE

Standard (Part 3)

Scope: Geometry, Trigonometry, and Probability Logic
Target: All SS1 Students (Questions 41 - 60)

Section C: Geometry & Trigonometry

No.	Question & Options	Ans	Diagnostic Focus
41	What is the sum of the interior angles of a hexagon? A) 360° B) 540° C) 720° D) 900°	C	Polygons (Sum of angles formula).
42	A ladder 10m long leans against a wall, making an angle of 60° with the ground. How high up the wall does it reach? A) 5m B) $5\sqrt{3}\text{m}$ C) $10\sqrt{3}\text{m}$ D) 8.66m	D	Applied Trigonometry (SOHCAHTOA).
43	Calculate the volume of a cylinder with radius 3cm and height 7cm. (Use $\pi = \frac{22}{7}$) A) 66cm^3 B) 132cm^3 C) 198cm^3 D) 462cm^3	C	3D Mensuration (Volume).
44	In a right-angled triangle, if the hypotenuse is 13cm and one side is 5cm, find the third side. A) 8cm B) 12cm C) 10cm D) 15cm	B	Pythagoras Theorem.
45	The angle of elevation of the top of a tower from a point 20m away on level ground is 30° . Find the height of the tower. A) $20\sqrt{3}\text{m}$ B) $\frac{20}{\sqrt{3}}\text{m}$ C) 10m D) 40m	B	Trigonometry (Angles of elevation).
46	Find the length of an arc that subtends an angle of 60° at the center of a circle of radius 21cm. A) 22cm B) 11cm C) 44cm D) 66cm	A	Circle Geometry (Arc length).
47	Which of these transformations preserves shape but changes size? A) Reflection B) Rotation C) Enlargement D) Translation	C	Transformational Geometry.
48	The bearing of point A from B is 045° . What is the bearing of B from A? A) 045° B) 135° C) 225° D) 315°	C	Bearings and Vectors.
49	Find the area of a triangle with sides 6cm and 8cm and an included angle of 30° . A) 24cm^2 B) 12cm^2 C) 48cm^2 D) 6cm^2	B	Trigonometric Area Formula.
50	If the scale factor of an enlargement is 3, what is the ratio of their areas? A) 1:3 B) 1:6 C) 1:9 D) 3:1	C	Similarity & Scale Factors.

Section D: Data, Logic & Probability

No.	Question & Options	Ans	Diagnostic Focus
51	Find the median of the following data: 5, 2, 9, 1, 5, 8, 3. A) 1 B) 5 C) 3 D) 9	B	Statistics (Central tendency).
52		A	Compound Probability.

No.	Question & Options	Ans	Diagnostic Focus
	Two fair dice are thrown. What is the probability that the sum is 7? A) $\frac{1}{6}$ B) $\frac{1}{12}$ C) $\frac{1}{36}$ D) $\frac{5}{36}$		
53	Which of the following is a measure of dispersion? A) Mean B) Mode C) Standing Deviation D) Median	C	Statistical Theory.
54	In a Venn diagram, $(\text{A} \cup \text{B})'$ represents: A) Elements in A and B B) Elements in A only C) Elements outside both A and B D) Elements in B only	C	Set Notation & Logic.
55	Calculate the standard deviation of a set of numbers where all values are the same. A) 1 B) 0 C) Undefined D) The mean	B	Conceptual Statistics.
56	If $\log_y 81 = 4$, find y . A) 3 B) 9 C) 2 D) 4	A	Logarithmic Foundations.
57	Express 0.00345 to two significant figures. A) 0.003 B) 0.0034 C) 0.0035 D) 0.00345	C	Accuracy and Approximations.
58	A sector of a circle has an area of 77cm^2 . If the radius is 7cm, find the angle at the center. A) 90° B) 180° C) 60° D) 45°	B	Reverse Mensuration.
59	Solve: $3x - (x - 2) = 10$. A) $x=4$ B) $x=6$ C) $x=3$ D) $x=5$	A	Algebraic grouping errors check.
60	Diagnostic Insight: Difficulty with Q42, Q48, and Q52 signals a student faces challenges with spatial reasoning and abstract probability, common in IGCSE.	-	Final Pathway determination.